SYSDESIGN REFERENCE

Microservices vs Monolith

Core concepts, reference diagram, tradeoffs, and interview prompts.

Choose service boundaries only when organizational and scaling benefits exceed distributed-system costs.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / MICROSERVICES-VS-MONOLITH

SYSDESIGN

Microservices vs Monolith

CORE IDEA

Choose service boundaries only when organizational and scaling benefits exceed distributed-system costs.

Reference architecture decision chain

01	02	03	04
Ownership	Deployment Cadence	Data Boundaries	Failure Isolation

Decisions to defend

+ Ownership
+ deployment cadence
+ data boundaries
+ failure isolation
+ and observability.

Failure modes to test

+ Chatty calls
+ shared databases
+ operational sprawl
+ and premature decomposition.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Microservices vs Monolith

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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